

Release Notes from Zhiqiang Jiang

EchoSight-Pro latest version 0.5.1

EchoSight-Pro version 0.5.1

2025-Sep-29th

Important algorithm change, artifacts in integrated GPU setting are optimized;
Research interface module added in for personalized academic research needs;

EchoSight-Pro version 0.5.0

2025-Sep-24th

Deep algorithm and system engineering optimization on Intel integrated GPU settings;
The fastest integrated GPU based synthetic, RTB software beam forming version by now;

EchoSight-Pro version 0.4.0

2025-Sep-10th

A lot of huge updates for integrated GPU settings;
Fixed sound field bugs in multiple MLA CPU setting;
Dynamic package size supported for both CPU and integrated GPU settings;

EchoSight-Pro version 0.3.0

2025-Aug-08th

Integrated Graphic Card setting added in and image correct checked on intel UHD 730;
Provide setting switch between CPU and Integrated GPU since this version, detailed information, plz check the user guidance manual;

```
<ACTIVE_TASK_CHAIN ECHO_SIGHT_PRO_MODE="OFF_LINE">CL_CHAIN</ACTIVE_TASK_CHAIN>  
<ACTIVE_TASK_CHAIN ECHO_SIGHT_PRO_MODE="OFF_LINE">CPU_CHAIN</ACTIVE_TASK_CHAIN>
```

EchoSight-Pro version 0.2.1

2025-July-25th

Independent GPU CUDA setting workable since this version;
CPU settings deep fine optimization done for extreme efficiency;
(For pure CPU computation software beaming, I don't know the future cases, but I am quite confidence it is the fastest engineering version on the planet of earth by now.)

EchoSight-Pro version 0.2.0

2025-July-15th

Provide standard/flash version both with identical output
Aggressive policy for flash version, in order to get higher performance (risks ???)
Some efficiency improvements;
Structure change to unify GPU and CPU;
Small bug fix;
Pre/Post gain configuration provided;

EchoSight-Pro version 0.1.2

Error fix in large MLA case;
Calculation modification updated for efficiency;
Overall performance for CPU only mode improves;
Full chain data rate maximum limit breakthrough;

EchoSight-Pro version 0.1.1

Basic calculation structure changed;
Unified data structure for CPU and GPU;
For CPU mode, 1MLA basic case efficiency will improve, but multi-MLA will drop a little, this is some sacrifice for upcoming GPU performance;

EchoSight-Pro version 0.1.0

Some big change in calculation updated;
Some performance improvements;
Some bugs fixed;

EchoSight-Pro version 0.0.3

MLA and Synthetic supported for linear probe in CPU only mode;
Focus calculation updated;
Sound field calculation updated;
Basic gain control model added in;
Some bugs within beam forming fixed;

EchoSight-Pro version 0.0.2

Huge milestone for EchoSight-Pro.
Linear probe workable and achieved amazing performance in new framework. The complete chain processing data rate on CPU only settings has defeated old framework EchoSight GPU only mode (old EchoSight performance at early-stage) already!

EchoSight-Pro version 0.0.1

Initial version tag of EchoSight-Pro.
The whole complete rewrite of the framework, data chain initial workable first-time status. No release to public.